

YANG ZHANG

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PROFILE

I'm currently a Ph.D. student at École Polytechnique Fédérale de Lausanne, under the supervision of Prof. Zolt Patakfalvi. My interest is in algebraic geometry in positive characteristics.

EDUCATION

Mathematics, Ph.D. École Polytechnique Fédérale de Lausanne	[Fall 2025, Now]
Mathematics, Exchange École Polytechnique Fédérale de Lausanne	[Fall 2024, Fall 2025]
Mathematics, Master of Science Universität Bonn	[Spring 2023, Fall 2025]
Mathematik, Bachelor of Science Universität Bonn	[Fall 2020, Spring 2023]

WRITINGS

A criterion for associated bundles and the weak Beauville–Bogomolov decomposition
arXiv:2608.10219

Automorphisms of odd dimensional $(2, 2)$ -complete intersections in characteristic 2
arXiv:2601.04109

On split coverings of Calabi-Yau threefolds in positive characteristics
My master thesis. Available on my homepage.

On the bounded negativity conjecture
My bachelor thesis. Available on my homepage.

TEACHING

Tutor of Algebra III - rings and fields École Polytechnique Fédérale de Lausanne, Lecturer: Dr. Leonid Monin	Spring 2026
Tutor of Algebra II - groups École Polytechnique Fédérale de Lausanne, Lecturer: Prof. Andrei Negut	Fall 2025
Tutor of Algebra I (Commutative Algebra) Universität Bonn, Lecturer: Prof. Dr. Daniel Huybrechts	Spring 2024
Tutor of Einführung in die Algebra (Introduction to the Algebra) Universität Bonn, Lecturer: Prof. Dr. Daniel Huybrechts	Fall 2023
Tutor of Algebraic Geometry II Universität Bonn, Lecturer: Dr. Gebhard Martin	Spring 2023
Tutor of Algebraic Geometry I Universität Bonn, Lecturer: Dr. Gebhard Martin	Fall 2022
Tutor of Mathematik für Physiker (Mathematics for phycists) II Universität Bonn, Lecturer: Dr. Illia Karabash	Spring 2022

SEMINAR TALKS

Kunz's Theorem

Fall 2024

Technische Universität München

I presented an alternative proof of the Kunz's theorem on the regularity in positive characteristics using perfection. The proof is given by Bhatt and Scholze.

K-Stability

Spring 2024

Universität Bonn

My talk involves the construction of the moduli stack of K-semistable Fano varieties by discussing their boundedness and openness.

Compactified Moduli of K3

Spring 2024

Universität Bonn

I gave the first talk which discusses the basic properties of complex K3 surfaces and moduli spaces of polarized K3 of low degrees.

Chow Groups and Motives

Fall 2022

Universität Bonn

The content of my talk consists of introducing the intersection product on the chow ring of a variety, proof of the moving lemma and the projection formula, and the compatibility of intersection product and cup product.

Klassische Algebraische Geometrie(Classical Algebraic Geometry)

Fall 2021

Universität Bonn

I gave a half part in each of two talks. The first talk is about defining product of two projective spaces using the Segre map. The second talk presents the classification of cubic curves on the projective plane.

DISTINCTIONS

IMC Maths 2023, Bronze medal.

IMC Maths 2022, Silver medal.

NOIP China 2018, first award.

EXTRAS

Language skills

Chinese (native), German (C1+), English (C1+), French (A2).

Accountant of student company AMBIX

[Fall 2017, Fall 2018)

I was active in AMBIX, a student party focusing on organizing business imitation contest for middle school students in grade 9 to 12.